Michael ("Mike") Hyland

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SYSTEMATIC TRADER & DERIVATIVES PORTFOLIO MANAGER

Former Global Head of Derivatives Trading with 15+ years of institutional experience as a market-maker and multi-asset portfolio manager. Currently developing and trading systematic relative-value strategies across crypto futures and options. Expertise in volatility modeling, market microstructure, and risk-adjusted portfolio construction. Has designed and deployed real-time market data ingestion, signal generation, and automated execution code in Python.

INSTITUTIONAL TRADING EXPERIENCE

BNY Mellon | Global Head, OTC Multi-Asset Derivatives Trading

- Led global trading organization (30+ traders, quants, technologists) across NY, London, Tokyo, and Hong Kong
- Managed **$3B+ balance sheet** across rates, FX, and equity derivatives (was 7th largest by size among US commercial banks according to OCC)
- Delivered **$100MM+ annual revenues** during 2008–2009 market stress
- Built and deployed early **XVA framework** for CVA and capital pricing
- Generated **~$40MM** through systematic arbitrage of interbank CVA and discounting inconsistencies
- Implemented real-time risk monitoring, pricing automation, and trade capture enhancements

SIG / BNY Derivatives Joint Venture | OTC Rates Derivatives Market-Maker

- Founding member of joint-venture trading desk; built infrastructure and pricing tools from inception
- Trained in Susquehanna's options-driven market-making program
- Managed customer flow and hedged nonlinear risk using listed and OTC instruments
- Developed relative-value frameworks across:
  - Swaps, Treasuries, FRAs, and futures (including convexity effects)
  - Swaptions, bond options, futures options, and caps/floors
- Built volatility surface models (SABR, SVI, SIG's wow/alpha) and forward-vol decomposition tools
- Created analytics tools for pricing, risk measurement, and P&L attribution

FAMILY OFFICE TRADING EXAMPLES

Fixed Income ETF Relative Value Strategy (Live) | February – August 2020 (~ six months)

- Capital: ~$100K | Turnover: ~$30MM
- Return: 21% | **CAGR: 55%** | **Sharpe: 3.4** | Max Drawdown: -1.8% | t-stat: 2.3
- Captured liquidity dislocations during COVID through pairs trading of highly similar bond ETFs (GitHub here)

CME Crypto Calendar Spread Strategy (Live) | May 2024 – August 2025 (~ 15 months)

- Capital: ~$500K
- Performance improved over time:
  - Total period - CAGR: 8% | Sharpe: 1.4 | Max Drawdown: -6.5% | t-stat: 1.6
  - Last six months - **CAGR: 18% | Sharpe: 4.0** | Max Drawdown: -0.7% | t-stat: 2.8
- Progressed from standard carry strategy (ETFs vs futures) to more capital-efficient alternative that exploited mispricings in term structure across CME crypto futures (primarily four-week vs five-week spread tenors)
- Built Python framework for signal generation, risk control, and performance analysis (GitHub here)

RECENT AND CURRENT TRADING RESEARCH

BTC Buy-and-Hold Alternative Strategy (Research)

- Three-year training period (2021-2023) and two-year testing period (2024-2025)
- Developed long only alternative to BTC buy-and-hold strategy (GitHub here)
    - Buy/Sell signal based on extreme positive or negative spikes in short-term Sharpe-like ratios
    - Four alternative cryptos and a much smaller amount of BTC used as strategy's trading assets
- Strategy's performance stats were as good or better than BTC buy-and-hold (in parenthesis)
    - Test period - CAGR: 105% (59%) | Sharpe: 2.1 (1.0) | Max Drawdown: -19% (-37%) | **Annual Alpha: 29%**
    - Had superior Sharpe ratio on 92% of rolling annual periods during training and testing days combined

Cross-Venue Crypto Options Microstructure Research (Ongoing)

- Currently building real-time Python tools to compare crypto options across different venues (options on ETFs, options on futures, and native streaming options)
- Analyzing differences in products, settlement, equivalence ratios, carry rates, forward prices, and expiration timing to identify market-making and relative value opportunities (GitHub here)

CAREER CHRONOLOGY

PWC | Bank Trading Consultant | 2022–Present

- Provide subject-matter expertise on trading and risk management related issues for large institutions
- Support projects involving model validation, capital markets risk, and front-office process design

Independent Derivatives Consultant and Stay-at-Home Dad | 2011–2021

- Expert advisor on derivatives trading and valuation in major litigation and regulatory matters
- Provided analysis related to LIBOR rigging probe, Lehman derivatives valuations, and trading fraud cases
- Joined my wife in providing focused support to our three children as they matured into fully contributing members of society

BNY Mellon | Global Head, OTC Multi-Asset Derivatives Trading | 2007–2010

- See first page

SIG / BNY Derivatives Joint Venture | OTC Rates Derivatives Market Maker | 1997–2006

- See first page

JP Morgan | Account Representative | 1989–1992

- Managed corporate ADR relationships and completed management training program

EDUCATION

NYU Stern School of Business | 1990-1993; MBA, Finance (with Distinction)
University of Pennsylvania | 1986-1989; B.A., Computer Mathematics, cum laude

TECHNICAL SKILLS

- Python (Pandas, NumPy, SciPy, asyncio)
- Volatility modeling: SABR, SVI, forward vol decomposition
- Market data integration: IBKR, crypto exchanges (WebSocket feeds)
- Strategy development, backtesting, performance analytics and portfolio construction
- Derivatives pricing, risk management, non-linear hedging and P&L attribution